

PROBABILITY THEORY

Session 2

Problem 1

The distribution function of the random variable X is given

$$F(x) = \begin{cases} 0 & x < 0 \\ x/2 & 0 \leq x < 1 \\ 2/3 & 1 \leq x < 2 \\ 11/12 & 2 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

- (a) Plot this distribution function.
- (b) what is $P\{X > 1/2\}$?
- (c) what is $P\{2 < X \leq 4\}$?
- (d) what is $P\{X < 3\}$?
- (e) what is $P\{X = 1\}$?

Problem 2

The amount of time, in hours, that a computer functions before breaking down is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} \lambda e^{-x/100} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

What is the probability that a computer will function between 50 and 150 hours before breaking down? What is the probability that it will function less than 100 hours?

Problem 3

Let X be the number of the cars being repaired at a repair shop. We have the following information:

- (a) At any time, there are at most 3 cars being repaired.
- (b) The probability of having 2 cars at the shop is the same as the probability of having one car.
- (c) The probability of having no car at the shop is the same as the probability of having 3 cars.

- (d) The probability of having 1 or 2 cars is half of the probability of having 0 or 3 cars.

Find the PMF of X .

Problem 4

The joint probability density function of X and Y is

$$f(x, y) = \begin{cases} xe^{-(x+y)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Compute the density of X .
 (b) Compute the density of Y .
 (c) Are X and Y independent?

Problem 5

If the probability density function of X equals

$$f(x) = \begin{cases} ce^{-2x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Find c . What is $P\{X > 2\}$?

Problem 6

A message is transmitted from A to B. Probability to receive a message is 0.8. A message is retransmitted until it is successfully received. Let X be a random variable represented a number of times it took to send the message until it is successfully received. Find pmf of X .

Problem 7

Let X represent the difference between the number of heads and tails obtained when a coin is tossed n times. What are the possible values of X ?

Problem 8

We roll two dice and observe two numbers X and Y .

- (a) Find marginal and joint pmfs of X and Y .
 (b) Find $P(X = 2, Y = 6)$.
 (c) Find $P(X > 3 | Y = 2)$.